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Mining resource profiles from event logs.

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Mining Resource Profiles From Event Logs

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In most business processes several activities need to be executed by human resources and cannot be fully automated. In order to evaluate resource performance and identify best practices as well as opportunities for improvement, managers need objective information about resource behaviours. Companies often use information systems to support their processes and these systems record information about process execution in event logs. We present *a framework for analysing and evaluating resource behaviour through mining such event logs*. The framework provides: (1) a method for extracting descriptive information about resource skills, utilisation, preferences, productivity and collaboration patterns; (2) a method for analysing relationships between different resource behaviours and outcomes; and (3) a method for evaluating the overall resource productivity, tracking its changes over time and comparing it with the productivity of other resources. To demonstrate the applicability of our framework, we apply it to analyse employee behaviour in an Australian company and we evaluate its usefulness by a survey among industry managers.

CCS Concepts: • **Information systems** → Information systems applications; • **Information systems applications** → Decision support systems;

General Terms: Performance, Measurement, Algorithms, Human Factors

Additional Key Words and Phrases: resource profile, event log, mining resource behaviour, evidence-based performance evaluation

1. INTRODUCTION

Employees are almost always involved in the execution of business processes and they play a fundamental role in the creation of value in an organisation. In order to better understand areas of improvement of resource¹ and process performance, managers need objective information about the working behaviours of teams and individual employees. Often managers do not have such information and they have to make decisions based on their subjective judgments [Manoharan et al. 2009]. This may result in inadequate performance appraisal systems, ineffective employee development strategies, and missed opportunities for performance improvement. For example, some teams may be overloaded or important activities may be assigned to inefficient resources. Identification of such patterns of resource behaviour can help to improve individual employee and team performance and overall process performance.

Given this background, the research question that emerges is: *How can resource behaviour be analysed in an objective way?* Business processes are often supported

¹In this article we use the term *resource* as a generic term for human resources; it can refer to an individual employee or a group of employees, e.g. a role or a team.

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by information systems which record information about process executions in event logs. Although resources play an important role in process executions, there are few methods currently available that allow organisations to extract knowledge about resource behaviour from event logs, with some notable exceptions [Song and van der Aalst 2008; van der Aalst et al. 2005; Nakatumba and van der Aalst 2010; Ly et al. 2006], but these focus on organisational structures and do not consider changes in resource behaviour over time (with the exception of the work of Nakatumba and van der Aalst [Nakatumba and van der Aalst 2010] in which changes in resource workload are considered). In a dynamic world, resources, as well as other process artefacts, change and “it is important to be able to get” their “documented history” [Beheshti et al. 2016]. For example, the influence of contextual factors (e.g. holiday season, weather) on the efficiency of resources can be observed over time [Leyer 2011]. In addition, resource planning can be described as a dynamic decision system that is characterized by delays between actions and results [Sterman 1989]. If a new policy to guide resources in their work is set up, it is very likely that the effect in terms of efficiency can be observed with a delay but the time of the delay is typically unknown. Thus, we present a framework for analysing resource behaviour through event log mining which allows tracking changes over time. Figure 1 depicts the main idea of our framework.

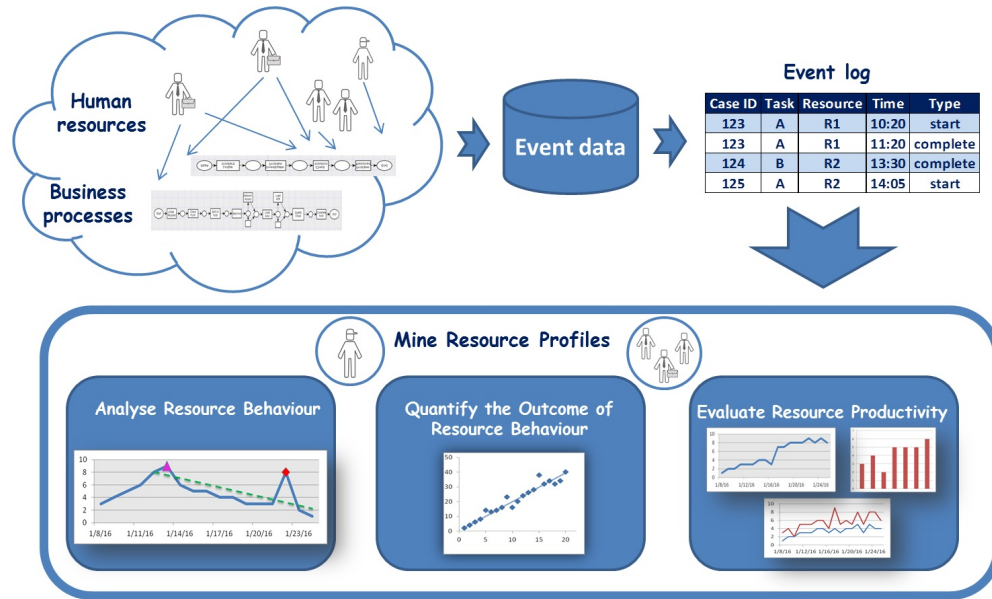


Fig. 1. Mining resource profiles (team or individual) from event logs.

Analysing resource behaviour can be seen from different perspectives with measures being structured according to the most relevant aspects [de Leeuw and van den Berg 2011]. As we focus our analysis on the actors (resources) involved in undertaking activities within a process, three perspectives appear to be relevant [Sawhney and Chason 2005; White et al. 1999; de Leeuw and van den Berg 2011]:

- The resource actions, i.e. what the resource has been doing? This is a descriptive view of resource behaviour. This leads to Research Question 1: *How can the behaviours of resources be identified?*

- The effect of resource behaviours. This covers the relationship between resource behaviours and outcomes and is thus a more analytic perspective. This leads to Research Question 2: *How can the effects of resource behaviours be quantified?*
- Evaluation and comparison of resource productivity. Here, a comparison is made with other resources over different time periods. This leads to Research Question 3: *How can the overall productivity of resources be evaluated?*

All three questions rely on the same input data, but take different perspectives in analysing resource behaviour. As such the analyses can be performed independently of each other. While the results are related from a content point of view, the separation into the three perspectives allows a clear separation of the specific questions of interest and acknowledges that different methods are required to address the questions.

In our earlier work [Pika et al. 2014] we presented a framework which allowed us to answer the first question by analysing *Resource Behaviour Indicators* (RBIs) in the following categories of resource behaviour: skills, utilisation, preferences, productivity and collaboration. However, it did not allow us to investigate the effects of this behaviour and it did not provide a means for evaluating overall resource productivity. We present in this article an extended framework that includes a new method for investigating whether or not any relationships exist between given resource behaviours and outcomes by analysing an event log (e.g. whether or not employees complete more work when they multi-task) (Section 3.3). The proposed method is based on regression analysis and allows us to investigate relationships from the following three process perspectives: *case*, *task* and *time*. Another technical contribution of this paper is a new method for evaluating resource productivity by analysing event logs (Section 3.4). The method uses the Data Envelopment Analysis (DEA) technique [Bogetoft and Otto 2011] for this purpose. It allows us to compare the productivity of different resources (teams or individual employees) and to track productivity evolution over time. The framework is based on a set of pre-defined measures and it also allows users to define their own indicators. We present the results of a new case study in which our framework was applied to analyse resource behaviours in an Australian company (Section 4.1) and the results of a new survey that evaluates the usefulness of the framework among managers (Section 4.2).

2. RELATED AND PREVIOUS WORK

In this section we first introduce business processes and human resource behaviour, then discuss process mining focusing on techniques for extracting human resource behaviour, and we conclude the section with a discussion of resource benchmarking.

Business processes. A business process is a coordinated sequence of activities that jointly realise a business goal and which are performed “in an organizational and technical environment” [Weske 2007]. People are considered a core element of Business Process Management (BPM) [Rosemann and vom Brocke 2015]. Business processes whose executions depend on knowledge workers performing knowledge-intensive tasks are referred to as knowledge-intensive processes [Ciccio et al. 2015]. Knowledge-intensive business services “have grown exponentially over the last three decades” [Freel 2016] and “have been the most dynamic component” in developed economies [Wirtz and Lovelock 2016]. Despite knowledge workers playing a prominent role in modern organisations, the BPM community traditionally provided “minor emphasis of collaboration aspects” [Ciccio et al. 2015].

Human resource behaviour. Process outcomes depend on human resource behaviour which is considered to be the “most important element that can affect project success” [Thevendran and Mawdesley 2004]. Human behaviour can be complex as peo-

ple “seek to promote their utility, through advancing their interests, preferences and ideas” and their decisions are influenced by their social contexts [Leftwich 2015]. Resource behaviour can also evolve over time [Beheshti et al. 2016]. Hence, human performance planning and measurement can become a challenging task, which is typically handled by human resource management departments [Espinilla et al. 2013; Peretz and Fried 2012]. Traditional performance evaluation approaches often suffer from “lack of objectivity, prejudice or halo errors” [Espinilla et al. 2013] and they often have a “narrow, or uni-dimensional, focus” [Neely et al. 2000]. One way to overcome the lack of objectivity is to base performance evaluation on “the opinion of different groups of reviewers who socialize with evaluated employees” [Espinilla et al. 2013]. Various authors have proposed performance evaluation frameworks that are based on “a balanced set of measures”; they “suggest some areas in which measures of performance might be useful, but provide little guidance on how the appropriate measures can be identified” [Neely et al. 2000]. Such performance measures can be defined on an aggregate level, namely, for teams, departments or a company [Nudurupati et al. 2011] or for individual employees [Neely et al. 2005; Thompson and Goodale 2006; Dulebohn and Johnson 2013].

Process mining. Business processes are not always executed as expected “due to the high variability that may affect operational processes in real word scenarios” [Ceravolo et al. 2016]. Processes are often supported by information systems which record information about their executions in event logs [van der Aalst 2016; Ceravolo et al. 2016]. Process mining aims to extract knowledge about business processes from such event logs [van der Aalst 2016]. Early process mining techniques were aimed at discovering process models from event logs; later on, process mining algorithms were developed that also support the discovery of other process perspectives (e.g. organisational, case or time perspectives) [van der Aalst 2016]. Some aspects of our framework were inspired by existing process mining approaches. In our own earlier work we presented process risk indicators for the identification of case delays [Pika et al. 2013a; Pika et al. 2013b]. We showed that a high resource workload or the involvement of certain resources in a case can contribute to case delays. De Leoni et al. [de Leoni et al. 2014] proposed a general framework for correlating process characteristics linked to events by applying decision tree analysis. Van der Aalst [van der Aalst 2013] proposed the notion of process cubes whereby events can be “organized using different dimensions” such as time windows or event or case types. Bose et al. [Bose et al. 2013] proposed a framework for detecting changes in processes. Pika et al. [Pika et al. 2016] presented a method for evaluating and predicting overall process risk which considers process evolution. However, the above-mentioned approaches focus on business processes rather than resources.

Extracting human resource behaviour. Several methods for extracting knowledge about some aspects of resource behaviour from event logs have been proposed in the process mining field: Song et al. [Song and van der Aalst 2008] proposed techniques for extracting organisational models; Van der Aalst et al. [Song and van der Aalst 2008; van der Aalst et al. 2005] presented a method for extracting social networks from event logs; Huang et al. [Huang et al. 2012] proposed some measures for resource preference, availability, competence and cooperation; and Nakatumba et al. [Nakatumba and van der Aalst 2010] presented a method that investigates the effect of resource workload on service times. Several approaches have been devised that use event logs to derive resource allocation mechanisms: Liu et al. [Liu et al. 2012] proposed an approach that mines resource allocation rules from event logs by applying a data mining algorithm; Cabanillas et al. [Cabanillas et al. 2013] proposed an approach for prioritising potential task performers based on their preferences; Kumar et al. [Kumar et al.

2013] devised a model that considers compatibility between resources when assigning tasks and a technique for learning resource compatibility from logs; and Ly et al. [Ly et al. 2006] presented a method for mining task assignment rules based on decision tree learning. These approaches focus on organisational structures, few specific measures or resource allocation mechanisms (rather than different aspects of employee behaviour) and they do not consider changes of resource behaviour over time.

Resource benchmarking. We adopted DEA [Bogetoft and Otto 2011] to evaluate and compare resource productivity. DEA is a popular efficiency evaluation and benchmarking technique based on linear programming that is often used in operations research to compare the efficiency of companies [Bogetoft and Otto 2011], departments or company functions [Thanassoulis 1995; Tavakoli and Shirouyehzad 2013] and business processes [Dohmen and Leyer 2010]. A few papers describe case studies in which DEA was applied to measure the efficiency of employees [Manoharan et al. 2009; Wagner et al. 2003; Koch-Rogge et al. 2014]. However, these case studies did not use event log data, they used data that was collected from different sources, manually pre-processed and transformed into a form suitable for DEA analysis.

3. FRAMEWORK

Our framework consists of three modules, as depicted in Figure 1, with each module tackling the three research questions introduced in Section 1. The goal of the first module, called *Analysing Resource Behaviour*, is to discover what resources have been doing, that is, to gain objective information about their skills, utilisation, working preferences, productivity and collaboration patterns (Research Question 1 in Section 1). The goal of the second module, called *Quantifying the Outcome of Resource Behaviour*, is to help managers to better understand the effects of resource behaviours on different process outcomes; for example, to check if the resource workload is affecting the duration of tasks executed by the resource or the quality of the work (Research Question 2 in Section 1). Our aim is *not* to learn whether or not certain resource behaviours are associated with particular outcomes. Instead, we aim to provide a method and a supporting tool that allows managers to observe what relationships exist between given resource behaviours and outcomes. Finally, the goal of the *Evaluating Resource Productivity* module is to provide a method for evaluating the overall productivity of a resource, by comparing it with the productivity of other resources and tracking its evolution over time (Research Question 3 in Section 1). Resources are often involved in multiple activities with different levels of complexity. Looking separately at each task would be prohibitively time consuming. Managers would benefit from a method that can automatically evaluate the productivity of a resource considering different resource inputs and outputs. They can further explore some resources of interest (e.g. over-performing or under-performing) and look at individual indicators to investigate resource behaviours and their effects in detail (using other modules of the framework).

Figure 2 depicts inputs and outputs of our framework. An input to the three modules is an event log which we discuss in detail in Section 3.1. Another input to the framework are resource behaviour measures. These are different for the three modules and we discuss them in Sections 3.2-3.4. We propose a number of measures of resource behaviour and also provide an interface that allows users to define their own measures. Outputs of the framework are three different types of analysis visualised by corresponding charts. These will be discussed in Sections 3.2-3.4. In the following sections, we first describe definitions of the basic concepts that are used in the framework specifications. We then describe in detail the three modules of our framework.

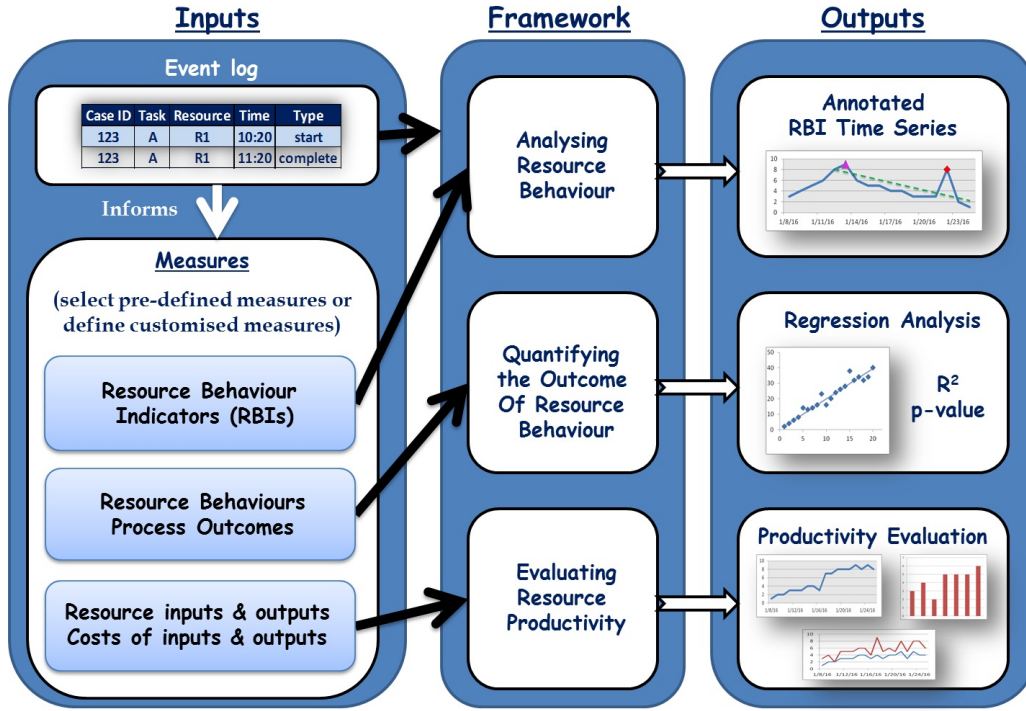


Fig. 2. Inputs and outputs of the framework for mining resource profiles from event logs

3.1. Definitions

A key input to our framework is an event log that captures information about process executions². Let $\mathcal{E}$ be the set of all events, an *event log* EL is a set of events, i.e. $EL \subseteq \mathcal{E}$. Events can have different attributes. We assume that each event has at least the following attributes (referred to here as *basic* attributes): *caseid*, *task*, *type*, *time* and *resource*. We assume that *start* and *complete* event types are always recorded, and other event types (e.g. *create*) may also be recorded. We can treat an event log as a relation whose relation scheme is specified by the set of event attributes.

Let $\mathcal{C}$ be the set of all cases, a *case log* CL is a set of cases, i.e. $CL \subseteq \mathcal{C}$. Similarly, cases are characterised by different attributes. A case is uniquely identifiable by the case attribute *caseid*. A case log can also be treated as a relation whose relation scheme is specified by the set of case attributes.

The value of attribute a of case c is denoted as c_a while e_a denotes the value of attribute a of event e . For example, e_{caseid} is the case identifier of event $e \in \mathcal{E}$.

Events and cases can have other attributes (e.g. *cost* or *outcome*). As examples of such non-basic attributes, we use the event attributes *sum* and *creator* and the case attributes *case_type* and *feedback*.

Figure 3 depicts an example of an event log that contains five events with five basic attributes and attribute *case_type*. We can see, for example, that in an urgent case with identifier “111” Anne started task “Request Information” at 10:15 on 10/02/2006. Note that in this example attribute *resource* refers to individual employees, but it can also refer to a role or a team.

²The framework implementation takes as input an XES event log, a standard format for event log data, <http://www.xes-standard.org/>

caseid	task	time	type	resource	case_type
111	Investigate Incident	10/02/2006 10:00	start	Anne	urgent
111	Request Information	10/02/2006 10:15	start	Anne	urgent
112	Send Bill	11/02/2006 15:00	complete	Mike	normal
111	Investigate Incident	13/02/2006 13:45	complete	Anne	urgent
112	Update record	13/02/2006 11:37	start	Mike	normal

Fig. 3. Example of an event log.

We also derive from the basic event log the case attributes *case_duration* (the time difference between the time stamps of the last and first event of a case) and *case_resources* (the number of resources that were involved in a case). The following event attributes are derived from the basic event log: *task_duration* (the time difference between the corresponding activity's³ *complete* and *start* events), *workload* (the number of task instances that were started but not completed by a resource involved in an event before the moment of event execution), and *workload_duration* (the time period during which the resource's workload did not change). For example, if a resource starts executing task *a* at time t_1 and then starts another task at time t_2 and task *a* is not completed yet at t_2 , the resource's workload at time t_2 is 1 and the workload duration is $t_2 - t_1$. For each event we also create the attribute *eventid* which is a unique identifier of the event. If any of these derivable event or case attributes are recorded in a log, users may choose to use the log's values instead of deriving them from the basic attributes.

Let R be a set of resources, A be a set of activities, t_1 and t_2 be the beginning and the end of a given time slot and r be a given resource. Below we define the functions that are later used in definitions of our methods.

Events completed during a given timeslot $[t_1, t_2)$:

$$E_{CT}(t_1, t_2) \triangleq \{e \in EL \mid e_{time} \geq t_1 \wedge e_{time} < t_2 \wedge e_{type} = \text{'complete'}\}$$

Events in which a given resource was involved during a given timeslot:

$$E_{TR}(t_1, t_2, r) \triangleq \{e \in EL \mid e_{time} \geq t_1 \wedge e_{time} < t_2 \wedge e_{resource} = r\}$$

Events completed by a given resource during a given timeslot:

$$E_{CTR}(t_1, t_2, r) \triangleq E_{CT}(t_1, t_2) \cap E_{TR}(t_1, t_2, r)$$

Cases completed during a given timeslot:

$$C_{CT}(t_1, t_2) \triangleq \{c \in \mathcal{C} \mid \exists e \in E_{CT}(t_1, t_2)[e_{caseid} = c_{caseid}] \wedge \nexists e' \in EL[e'_{caseid} = c_{caseid} \wedge e'_{time} > t_2]\}$$

Cases in which a given resource was involved:

$$C_R(r) \triangleq \{c \in \mathcal{C} \mid \exists e \in EL[e_{caseid} = c_{caseid} \wedge e_{resource} = r]\}$$

Cases in which a given resource was involved during a given timeslot:

$$C_{TR}(t_1, t_2, r) \triangleq \{c \in \mathcal{C} \mid \exists e \in E_{TR}(t_1, t_2, r)[e_{caseid} = c_{caseid}]\}$$

3.2. Analysing Resource Behaviour

Our first goal is to be able to answer Research Question 1: What have resources been doing? To extract descriptive information about different aspects of resource behaviour, we follow the three main steps: (1) defining the Resource Behaviour Indicators (RBIs),

³In this article we use the terms *task* and *activity* as synonyms.

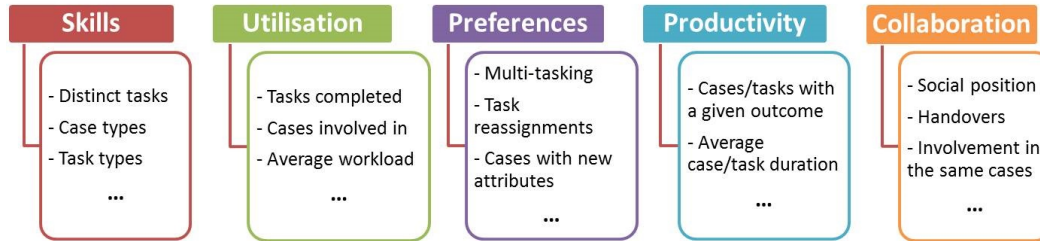


Fig. 4. Categories of resource behaviour and examples of RBIs

(2) extracting the RBI time series from the event logs, and (3) analysing the RBI time series.

3.2.1. Defining Resource Behaviour Indicators. The kinds of resource behaviour that managers may wish to analyse depend on the reasons for the analysis and the particular context. There may be many indicators of interest in a specific situation [Neely et al. 2005]. We refer to a collection of RBIs that are relevant in a particular context as a *resource profile*. Based on the literature we propose the following categories of resource behaviour [Pika et al. 2014]:

- (1) Skills [Thompson and Goodale 2006] (*What can a resource do?*)
- (2) Utilisation [Neely et al. 2005] (*What is a resource actually doing?*)
- (3) Preferences [Huang et al. 2012; van der Aalst 2010; van der Aalst 2015] (*What working behaviour does a resource often demonstrate?*)
- (4) Productivity [Murphy 1999] (*How good is a resource at what it⁴ does?*)
- (5) Collaboration [van der Aalst et al. 2005; Huang et al. 2012] (*How does a resource work with other resources?*)

We define a set of RBIs in each category (Figure 4). Many of our pre-defined RBIs can be extracted from basic event logs while others require additional information (e.g. cost or feedback) to be recorded. Furthermore, some RBIs are generic and others are only relevant in specific contexts; RBI values may be absolute or relative. We also provide an interface that allows users to define their own RBIs. For example, a manager may be interested in conformance and may wish to check whether a resource performs a task that should not be performed. In the remainder of this section we discuss RBIs in each of the categories of resource behaviour and provide formal definitions for a small selection of the RBIs. Let $RBI_n(t_1, t_2, r, [p_1 \dots p_n])$ denote the value of an RBI n during a given timeslot, t_1 to t_2 , for resource r and for an optional set of other parameters $p_1 \dots p_n$. For example, an optional parameter can refer to cost, customer feedback or product category.

1. Skills: What can a resource do? Resources have different capabilities and they acquire new skills at a different pace. Knowledge about resource capabilities can help in resource scheduling [Huang et al. 2012; Thompson and Goodale 2006] and resource development planning. We assume that a resource is capable of performing those types of activities it has performed in the past; hence, RBIs in this category reflect only “demonstrated” skills. Some example RBIs in this category include: the number of distinct tasks completed by a resource; the fraction of distinct tasks completed by a

⁴In this article we use the pronoun *it* when referring to resources as the term *resource* can refer to an employee or a group of employees (e.g. a role or a team).

resource with respect to the total number of distinct tasks completed; the number of completions of an activity with a given property by a resource; the fraction of completions of an activity with a given property by a resource with respect to the total number of completions of the activity with the given property; the number of case completions with a given property in which a resource was involved; the fraction of case completions with a given property in which a resource was involved with respect to the total number of case completions with the given property; and the fraction of completions of a given activity by a resource with respect to the total number of activity completions by the resource.

RBI 1.1 Distinct activities: The number of distinct activities completed by a given resource, r , during a given timeslot, t_1 to t_2 . This RBI is relevant in those working environments where resources learn new skills and get involved in more tasks over time.

$$\text{Distinct Activities}(t_1, t_2, r) \triangleq |\{task \in A \mid \exists e \in E_{CTR}(t_1, t_2, r)[e_{task} = task]\}|$$

RBI 1.2 Case types: The fraction of cases with a given value, p , of case attribute *case.type* completed during a given time slot, $[t_1, t_2)$, in which a given resource, r , was involved with respect to the number of cases completed during time slot $[t_1, t_2)$ in which the resource was involved (requires attribute *case.type* to be recorded in the log). Looking at the types of cases processed by a resource, managers can discover, for example, that the resource is only processing cases related to a specific product or that it is getting involved in more complex cases over time.

$$\text{Case.Types}(t_1, t_2, r, p) \triangleq \frac{|\{c \in C_{CT}(t_1, t_2) \cap C_R(r) \mid c_{case.type} = p\}|}{|C_{CT}(t_1, t_2) \cap C_R(r)|}$$

RBI 1.3 Activity frequency: The fraction of completions of a given activity, a , by a given resource, r , during a given timeslot, $[t_1, t_2)$, with respect to the total number of activity completions by resource r during timeslot $[t_1, t_2)$. Managers can discover, for example, those activities which a resource frequently executes (hence, the activities in which the resource is more experienced) and those that it executes only occasionally.

$$\text{Activity.Frequency}(t_1, t_2, r, a) \triangleq \frac{|\{e \in E_{CTR}(t_1, t_2, r) \mid e_{task} = a\}|}{|E_{CTR}(t_1, t_2, r)|}$$

2. Utilisation: *What is a resource actually doing?* Utilisation RBIs measure how active a resource is without looking into the quality of its outputs. These indicators are inspired by measures in manufacturing (e.g. the number of units produced) [Neely et al. 2005]. Users can look at the number of activity instances completed by a resource, the fraction of activity instances completed by a resource with respect to the total number of activity instances completed, the number of completions of a given activity by a resource, the number of completed cases in which a resource was involved, the fraction of completed cases in which a resource was involved with respect to the total number of cases completed, or the average resource workload.

RBI 2.1 Activity completions: The number of activity instances completed by a given resource during a given time slot.

$$\text{Activity.Completions}(t_1, t_2, r) \triangleq |E_{CTR}(t_1, t_2, r)|$$

RBI 2.2 Number of case completions: The number of cases completed during a given time slot in which a given resource was involved.

$$\text{Case.Completions.Number}(t_1, t_2, r) \triangleq |C_{CT}(t_1, t_2) \cap C_R(r)|$$

RBI 2.3 Fraction of case completions: The fraction of cases completed during a given time slot in which a given resource was involved with respect to the total number of cases completed during the time slot.

$$\text{Case_Completions_Fraction}(t_1, t_2, r) \triangleq |C_{CT}(t_1, t_2) \cap C_R(r)| / |C_{CT}(t_1, t_2)|$$

RBI 2.4 Average workload: The average number of activities started by a given resource but not completed at a moment in time. We do not consider the resource's working hours, hence this RBI only yields a rough estimation of the resource's average workload.

$$\text{Average_Workload}(t_1, t_2, r) \triangleq \frac{\sum_{e \in E_{TR}(t_1, t_2, r)} e_{\text{workload}} * e_{\text{workload_duration}}}{\sum_{e \in E_{TR}(t_1, t_2, r)} e_{\text{workload_duration}}}$$

3. Preferences: *What working behaviour does a resource often demonstrate?* Resources have different working styles that may affect their performance [van der Aalst 2010; van der Aalst 2015; Huang et al. 2012]. We may learn about resources' preferences by checking if they often multitask, execute only similar tasks or take risks by performing a multitude of tasks, perform more work during certain weekdays or reassign tasks to others.

RBI 3.1 Multi-tasking: The fraction of active time during which a given resource is involved in more than one activity with respect to the resource's active time. We do not consider the resource's working hours and we assume that the resource is working on all the tasks that they started. Hence, this RBI only yields a rough estimate of the resource's multitasking preference.

$$\text{Multitasking}(t_1, t_2, r) \triangleq \frac{\sum_{e \in E_{TR}(t_1, t_2, r), e_{\text{workload}} > 1} e_{\text{workload_duration}}}{\sum_{e \in E_{TR}(t_1, t_2, r), e_{\text{workload}} > 0} e_{\text{workload_duration}}}$$

RBI 3.2 New attribute values: The number of times a resource completed a task during a given timeslot with an attribute value never seen before for any event (e.g. for the attribute *sum*). This RBI reflects the resource's propensity to execute new and hence risky tasks. For the attribute *sum*, the RBI can be formalised as follows:

$$\text{New_Attribute_Values}(t_1, t_2, r, \text{sum}) \triangleq |\{e \in E_{CTR}(t_1, t_2, r) \mid \nexists e' \in EL [e'_{\text{sum}} = e_{\text{sum}} \wedge e'_{\text{time}} < e_{\text{time}}]\}|$$

RBI 3.3 Activity reassignments: The number of times when an activity started by a given resource during a given time slot was later completed by a different resource⁵.

$$\text{Activity_Reassignments}(t_1, t_2, r) \triangleq |\{e \in E_{TR}(t_1, t_2, r) \mid e_{\text{type}} = \text{start} \wedge \exists e' \in EL [e'_{\text{type}} = \text{complete} \wedge e'_{\text{resource}} \neq r \wedge e'_{\text{task}} = e_{\text{task}} \wedge e'_{\text{caseid}} = e_{\text{caseid}} \wedge e'_{\text{time}} > e_{\text{time}}]\}|$$

4. Productivity: *How good is a resource at what it does?* Productivity RBIs aim to measure a resource's results; for example, its results in terms of the timeliness, cost or quality of outputs (assuming this information is recorded in the event log). Here we define RBIs for: the number of activities/cases completed with a given outcome in which a resource was involved; the fraction of activities/cases completed with a given outcome in which a resource was involved with respect to the total number of activities/cases completed with a given outcome; the average value of a given outcome

⁵The definition is provided for cases without activity repetitions. If activities are repeated, one would need to use activity instance identifiers.

for cases/activities completed in which a given resource was involved; and the number of times when a given activity was repeated when completed by a resource.

RBI 4.1 Activity outcomes: The fraction of activities completed with a given outcome (e.g. *duration*) during a given time slot by a given resource with respect to the total number of activities completed by the resource during the time slot. For activity duration, the RBI can be formalised as follows:

$$\text{In_Time_Activities}(t_1, t_2, r, \text{dur}) \triangleq \frac{|\{e \in E_{CTR}(t_1, t_2, r) \mid e_{task.duration} < \text{dur}\}|}{|\{E_{CTR}(t_1, t_2, r)\}|}$$

RBI 4.2 Case outcomes: The fraction of cases completed during a given time slot with a given outcome (e.g. customer feedback *cf*) in which a given resource was involved with respect to the total number of cases completed during the time slot in which the resource was involved. This RBI requires a case outcome attribute to be recorded, for example *feedback* (a numeric value, e.g. a customer rating). For the attribute *feedback*, the RBI can be formalised as follows:

$$\text{Satisfactory_Cases}(t_1, t_2, r, cf) \triangleq \frac{|\{c \in C_{CT}(t_1, t_2) \cap C_R(r) \mid c_{feedback} \geq cf\}|}{|C_{CT}(t_1, t_2) \cap C_R(r)|}$$

RBI 4.3 Average duration of a given activity: The average duration of instances of a given activity completed during a given time slot by a given resource.

$$\text{Average_Activity_Duration}(t_1, t_2, r, a) \triangleq \frac{\sum_{e \in E_{CTR}(t_1, t_2, r), e_{task}=a} e_{task.duration}}{|\{e \in E_{CTR}(t_1, t_2, r) \mid e_{task} = a\}|}$$

RBI 4.4 Average case duration: The average duration of cases completed during a given time slot in which a given resource was involved.

$$\text{Average_Case_Duration}(t_1, t_2, r) \triangleq \frac{\sum_{c \in C_{CT}(t_1, t_2) \cap C_R(r)} c_{case.duration}}{|C_{CT}(t_1, t_2) \cap C_R(r)|}$$

RBI 4.5 Average customer feedback: The average customer feedback for cases completed during a given time slot in which a given resource was involved.

$$\text{Average_Customer_Feedback}(t_1, t_2, r) \triangleq \frac{\sum_{c \in C_{CT}(t_1, t_2) \cap C_R(r)} c_{feedback}}{|C_{CT}(t_1, t_2) \cap C_R(r)|}$$

5. Collaboration: *How well does a resource work with other resources?* It is important to measure the collaborative aspects of resource behaviour. RBIs in this category can help us to learn about a resource's collaboration patterns with some other resources (e.g. the number of handovers from/to a given resource or the number of times when two given resources were involved in the same cases), or to obtain insight into a resource's overall social position within an organisation (e.g. the number of other resources that executed a given activity, the average number of resources involved in the same cases with a given resource or the fraction of resources involved in the same cases with a given resource during a given time slot with respect to the total number of resources active during the time slot).

RBI 5.1 Interactions between two given resources: The number of cases completed during a given time slot in which two given resources were involved.

$$\text{Interactions_Between_Resources}(t_1, t_2, r, r_2) \triangleq |C_{CT}(t_1, t_2) \cap C_R(r) \cap C_R(r_2)|$$

RBI 5.2 Social position: The fraction of resources involved in the same cases with a given resource during a given time slot with respect to the total number of resources active during the time slot.

$$\text{Social_Position}(t_1, t_2, r) \triangleq \frac{|\{r_1 \in R \mid \exists c \in C_{TR}(t_1, t_2, r_1) \cap C_{TR}(t_1, t_2, r)\}|}{|\{r_1 \in R \mid \exists c \in C_{TR}(t_1, t_2, r_1)\}|}$$

RBI 5.3 Delegations: The number of times when a resource, r , assigns a task to another resource, r_2 . For this RBI, a corresponding attribute (e.g. *creator*) must be recorded in the log. For the attribute *creator*, the RBI can be formalised as follows:

$$\text{Delegations}(t_1, t_2, r, r_2) \triangleq |\{e \in E_{CTR}(t_1, t_2, r_2) \mid e_{creator} = r \wedge r \neq r_2\}|$$

3.2.2. Extracting RBI Time Series From an Event Log. In the second step we extract the RBI time series in order to track the evolution of a particular RBI over time. The RBI time series consists of RBI values extracted for a given period of time (e.g. per day or week) for a given resource. Let $RBI_n(t_1, t_2, r, [p_1 \dots p_n])$ denote the value of an RBI n during a given time slot, t_1 to t_2 , for a given resource, r , and optional parameters $p_1 \dots p_n$; TS_{start} be the starting time point; $TS_{slotsize}$ be the sampling rate; TS_{size} be the number of time slots; and $Start(t)$ and $End(t)$ be functions that return the beginning and the end of a timeslot for a given time t respectively. An RBI time series is defined as:

$$TS_{RBI_n} \triangleq \{(RBI_n(Start(t), End(t), r, [p_1 \dots p_n]), t) \mid t \in \{TS_{start} + i * TS_{slotsize} \mid i \in \{0, 1, \dots, TS_{size} - 1\}\}\}$$

Here we use the following functions for the beginning and the end of a time slot: $Start(t) = t$ and $End(t) = t + TS_{slotsize}$; other functions can be defined⁶. (This gives flexibility to use different time series sampling methods. For example, one may define overlapping time slots.) The starting time point for the analysis, the time series sampling rate and the number of time slots are the input parameters provided by a user. The time series sampling rate is an important parameter that can affect the analysis results. Selection of the time series sampling rate is a known problem often discussed in the literature [Lijffijt et al. 2012]. The time series sampling rate may depend on process characteristics (e.g. process granularity) and the type of analysis in which the user is interested. For example, if managers are interested in checking whether or not a team is less productive on Mondays, they will look at the daily RBI values. Figure 5 depicts an example of a time series (monthly values) for RBI 3.3 “Activity reassignments”. We can observe that some tasks were reassigned from resource *Anne* until April 2008, and that there were no subsequent reassignments.

3.2.3. Analysing the RBI Time Series. In the third step we analyse the extracted RBI time series and visualise the analysis results. Time series charts accompanied by trend lines can give many insights into the dynamics of resource behaviour; however, they are not very convenient when the amount of available data is large. To deal with this issue, we use automatic techniques for time series analysis such as the detection of outliers [van der Loo 2010] and change points [Ross and Adams 2012], and the time series comparison [Mann and Whitney 1947].

Many methods for change point detection have been developed; here we use non-parametric methods [Hawkins and Deng 2010; Ross and Adams 2012] that do not make assumptions about the distribution of the data. Users can choose non-parametric tests for detecting changes in location (Mann–Whitney), changes in scale (Mood) or arbitrary distributional changes (Kolmogorov–Smirnov, Cramer–von-Mises) [Ross and Adams 2012].

⁶ $Start(t) = t + a$ and $End(t) = t + TS_{slotsize} + b$, where a and b are input parameters.

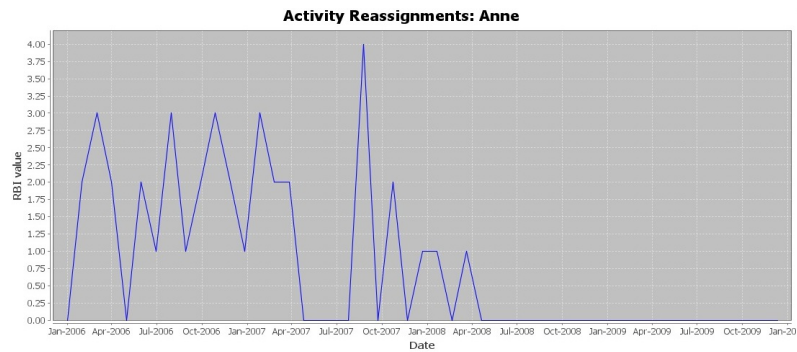


Fig. 5. Example of time series for RBI 3.3 “Activity reassignments”

Identifying outliers in the RBI time series (i.e. the points in time when RBI values are significantly different from typical values) can be helpful in the investigations of problems. We use an outlier detection method that first fits the distribution of the observations and then detects the observations that are unlikely to be generated by this distribution [van der Loo 2010].

Evaluation of the performance of a resource using RBIs can also be done by comparing the resource’s behaviour with the behaviour of other resources. This allows users to quickly identify those resources whose behaviour is different from others (e.g. under-performing or over-performing resources). To compare RBI time series we use the non-parametric Mann–Whitney U-test [Mann and Whitney 1947].

3.3. Quantifying the Outcome of Resource Behaviour

While the *Analysing Resource Behaviour* module described above allows us to extract descriptive information about different aspects of resource behaviour, the goal of the *Quantifying the Outcome of Resource Behaviour* module is to provide a method that allows managers to check whether or not a given resource’s behaviours affect given outcomes (Research Question 2 in Section 1). For example, a manager may be interested to learn how a resource’s workload affects the quality of their work, or if the case duration is influenced by the number of resources processing a case.

We use regression analysis to quantify the relationships between resource behaviours and outcomes. This is a statistical technique used for modelling the relationship between variables [Montgomery et al. 2012]. We use regression analysis because it is a popular technique often used for the investigation of relationships between different social and economic phenomena and its results are easy to interpret for business users. The framework supports linear regression analysis [Montgomery et al. 2012] as well as non-parametric kernel-based regression which does not make assumptions about data distribution [Racine and Li 2004].

To perform regression analysis it is necessary to define the dependent variable and one or more independent variables. Our framework provides an interface for users to define the dependent and independent variables they would like to analyse. It then extracts the values of the defined variables from an event log, fits a regression model and provides p – value, R^2 [Gujarati 2004] (coefficient of determination) and a plot of data and the fitted regression model (in the case of one independent variable).

In the context of business processes we can look at resource behaviours and outcomes that relate to: *cases* (e.g. the relationship between the percentage of tasks executed in a case by a given resource and the case duration or cost); *tasks* (e.g. the relationship between a task outcome and the experience of a resource executing the task); or we

can look at resource behaviours and outcomes during a given period of *time* (e.g. the relationship between the number of distinct tasks completed per week and the number of task instances completed during the week). Next, we describe how to analyse whether or not relationships exist between given resource behaviours and outcomes for the three perspectives referred to here as the *case*, *task* and *time* perspectives. The procedure consists of the following steps: 1) define a set of instances to be included in the analysis (i.e. cases, task instances or time slots); 2) define the value of the dependent variable for a given instance; and 3) define the value of an independent variable for a given instance.

Case perspective

- (1) Define a set of cases to be included in the analysis, $\mathcal{C}_{RA}$. The cases in which a user may be interested include all completed cases, cases in which a given resource was involved or cases of a certain type. For example, $\mathcal{C}_{RA}$ may comprise all cases in which a given resource, r , was involved: $\mathcal{C}_{RA} \triangleq \mathcal{C}_R(r)$.
- (2) Define the value of the dependent variable DV for a given case $c \in \mathcal{C}_{RA}$ (e.g. one may want to check whether some resource behaviour affects the duration or cost of cases). For example, $DV(c)$ may yield the duration of case c :

$$DV(c) \triangleq c_{case_duration}.$$
- (3) Define the value of an independent variable IV_i for a given case $c \in \mathcal{C}_{RA}$ (e.g. one may want to check whether the percentage of tasks completed by a given resource in the case or the number of resources involved in the case affects the case outcome). For example, $IV_i(c)$ may yield the number of resources involved in case c :

$$IV_i(c) \triangleq |\{r \in R \mid \exists e \in EL[e_{resource} = r \wedge e_{caseid} = c]\}|.$$

Task perspective

- (1) Define a set of events that should be analysed, $\mathcal{E}_{RA}$. A user may wish to include in the analysis the events related to completed tasks, instances of a given task or tasks completed by a given resource. For example, $\mathcal{E}_{RA}$ may comprise all the events related to completed instances of a given activity a :

$$\mathcal{E}_{RA} \triangleq \{e \in \mathcal{E} \mid e_{task} = a \wedge e_{type} = complete\}.$$
- (2) Define the value of the dependent variable DV for a given event $e \in \mathcal{E}_{RA}$. For example, $DV(e)$ may yield the task duration: $DV(e) \triangleq e_{task_duration}$.
- (3) Define the value of an independent variable IV_i for a given event $e \in \mathcal{E}_{RA}$ (e.g. the experience or workload of a resource that executed the related task). For example, $IV_i(e)$ may yield the resource's workload: $IV_i(e) \triangleq e_{workload}$.

Time perspective

- (1) Define time series parameters, namely, the starting time point for the analysis, sampling rate (e.g. a day or a week) and the number of time slots one is interested in. For a given time slot there can be only one value of the dependent variable, and one value of each independent variable.
- (2) Define the value of the dependent variable DV during a given time slot (t_1, t_2) (e.g. the number of task instances completed by a resource, or the fraction of cases with a given outcome completed in which a resource was involved with respect to the total number of cases completed in which the resource was involved). For example, a user may look at the average duration of a task completed by a given resource, r :

$$DV(t_1, t_2, r) \triangleq \sum_{e \in E_{CTR}(t_1, t_2, r)} e_{task_duration} / |E_{CTR}(t_1, t_2, r)|.$$

- (3) Define the value of an independent variable IV_i during a time slot (t_1, t_2) (e.g. the average resource workload). For example, a user may look at the number of distinct tasks completed by a resource r :

$$IV_i(t_1, t_2, r) \triangleq |\{a \in A \mid \exists e \in E_{CTR}(t_1, t_2, r)[e_{task} = a]\}|.$$

The framework then extracts from an event log the values of the defined dependent variable DV and all independent variables IV_i for all cases in C_r (for the *case* perspective), events in $\mathcal{E}_r$ (for the *task* perspective) or time slots (for the *time* perspective), for example, weekly values of the average duration of tasks completed by a resource and weekly values of the average workload of the resource. It runs regression analysis using the variable values and provides R^2 , p – *value*, the fitted regression model details (in the case of multiple independent variables) and a plot (when one independent variable is considered, an example is provided later in Figure 15) that allow us to identify whether or not a relationship exists between given resource behaviours and outcomes.

3.4. Evaluating Resource Productivity

The goal of the *Evaluating Resource Productivity* module is to provide a method for evaluating and comparing the productivity of human resources and tracking the evolution of productivity over time (Research Question 3 in Section 1). If resources only execute one task, use the same inputs needed to perform this task (e.g. information or materials), and their working hours are the same, then it is possible to compare the productivity of these resources by simply counting the number of task instances completed by each resource during a given time period. However, this scenario is not realistic because resources in modern organisations are often involved in multiple tasks with different levels of complexity.

To be able to evaluate resource productivity in such complex environments we use the Data Envelopment Analysis (DEA) [Bogetoft and Otto 2011] technique. DEA takes as input combinations of inputs and outputs that were previously observed and estimates the best practice (the “efficient frontier”) [Bogetoft and Otto 2011]. This is a non-parametric method based on linear programming that can consider multiple inputs and outputs and is often used in operations research to compare the efficiency of companies [Bogetoft and Otto 2011], departments or company functions [Thanassoulis 1995; Tavakoli and Shirouyehzad 2013] and business processes [Dohmen and Leyer 2010], with typical inputs being money, materials and people and typical outputs being units of production. DEA does not require knowledge of the relation between inputs and outputs and is not limited to a certain underlying stochastic function of the data [Dohmen and Leyer 2010; Cooper et al. 2011]. Thus, there are not many disadvantages compared to other efficiency evaluation methods like free disposal hull [Cooper et al. 2011]. It has been shown that it can be applied on a process level using process instances as decision making units without severe limitations [Burger and Moormann 2008] and has been applied to assess resource behaviour in general without using data from event logs [Koch-Rogge et al. 2014].

In many organisations the amount of output (e.g. the number of completions of a given task) that should be produced by a resource in a given role using a given input (e.g. working hours) is unknown. The first challenge is to be able to identify productivity “best practice” for the role from data. The DEA technique extracts from data the “set of combinations of input and output such that the input can actually produce the output” (referred to as the “technology”) and estimates the best practice for the technology (referred to as the “efficient frontier”) [Bogetoft and Otto 2011]. When estimating the efficient frontier, users may select one of the four DEA models based on their return-to-scale assumptions: constant returns to scale (*CRS*), decreasing returns to scale (*DRS*), increasing returns to scale (*IRS*) or variable returns to scale (*VRS*) [Bogetoft and Otto

2011]. By default, we use the *CRS* model. In the context of business processes, a resource's working hours would typically be considered as an input. It seems reasonable to expect that the amount of a resource's output should be proportional to the amount of the resource's working time. For example, the number of tasks completed by a resource who works 40 hours per week should be twice as high as the number of tasks completed by a resource who only works 20 hours per week.

DEA also allows us to assign costs to inputs and outputs (if they are available) and to calculate optimal revenue output, optimal cost input or optimal profit [Bogetoft and Otto 2011]. For example, a user may wish to assign costs to tasks with different levels of complexity and consider them when estimating productivity best practice.

After learning the efficient frontier, that is, the productivity best practice, we would like to measure the productivity of resources with respect to this frontier. DEA also allows us to measure efficiency as related to the efficient frontier [Bogetoft and Otto 2011]. The productivity score for a given resource during a given time period is a value from 0 to 1, with 1 being the maximum possible productivity. If a resource's productivity score during a given time period is 1, this means that the combination of the resource's inputs and outputs during this time period belongs to the efficient frontier.

Our framework allows users to define inputs and outputs (and their costs if they are known) for a given resource during a given time slot, estimate the efficient frontier using the defined inputs and outputs from an event log, and evaluate resource productivity as related to this frontier.

As the first step, users should define the values of the resource inputs and outputs for a given time slot. On many occasions, resources use the same inputs (such as tools or information) which they require to conduct their work. For such cases there is no need to define the inputs as they are constant. However, in some cases, the inputs are necessary. Resource inputs and outputs that can be handled by our framework depend on information recorded in an event log. Here we focus on resource inputs and outputs that can be extracted from a basic event log. We also provide examples of resource inputs and outputs that can be extracted from richer event logs, however this is a direction for future work as we explain in Section 5.2. Let $R_i(t_1, t_2, r, [p_1 \dots p_n])$ denote resource input and $R_o(t_1, t_2, r, [p_1 \dots p_n])$ denote resource output during a given timeslot, t_1 to t_2 , for resource r and for an optional set of other parameters $p_1 \dots p_n$, for example, for a given activity.

- *Resource inputs* that can be extracted from a basic event log include: 1) the amount of time a resource works on a process during a given time slot (it should be considered as an input if it varies for different resources or if a casual worker's hours vary from week to week); 2) the number of cases active during a given time slot (it should be considered as input if the number of active cases varies during different periods, so that productivity scores are not affected during periods with fewer cases). For example, if we consider as input active cases, resource input can be defined as: $R_i(t_1, t_2, r) \triangleq |\{c \in \mathcal{C} \mid \exists e \in EL[e_{caseid} = c_{caseid} \wedge e_{time} \geq t_1 \wedge e_{time} < t_2]\}|$. Resource skills and experience are examples of resource inputs that require a richer log.
- *Resource outputs* that can be extracted from a basic event log include: 1) the number of instances of a given task completed by a resource during a given time slot; 2) the number of cases completed during a given time slot in which a resource was involved; 3) the number of task instances completed by a resource during a given time slot. For example, if we consider as output task completions, resource output can be defined as: $R_o(t_1, t_2, r) \triangleq |E_{CTR}(t_1, t_2, r)|$. An example of resource output that requires a richer event log is the number of cases of a given type (e.g. for a given product or of a certain complexity) completed during a given time slot in which a resource was involved.

As the second step, the user selects which data should be used to estimate the efficient frontier. Our framework provides three options:

- (1) Consider inputs and outputs for one resource during different time periods. This way, the framework will estimate a “personal” efficient frontier. An assumption here is that the frontier does not change over time, though the productivity of a resource can change.
- (2) Use inputs and outputs for multiple resources during one time period. An assumption here is that the efficient frontier is the same for the resources but their individual productivity can be different; hence, the user should only consider resources that do the same type of work.
- (3) Use inputs and outputs for multiple resources during different time periods. An assumption here is that the efficient frontier is the same for the resources and it does not change over time, but the resources can have different levels of productivity that can change over time.

Our framework uses defined inputs and outputs and extracts from an event log a set of observed input-output combinations for selected resources during selected time periods. It also allows users to define costs of inputs and outputs if they are available. It then estimates the efficient frontier using the set of input-output values, calculates resource productivity as related to the frontier and produces a chart illustrating the productivity. Figure 6a shows an example of output for productivity evaluation for one resource during different time periods; Figure 6b shows an example of output for productivity evaluation for multiple resources during one time period; and an example of output for productivity evaluation for two resources during different time periods is provided later in Figure 16.

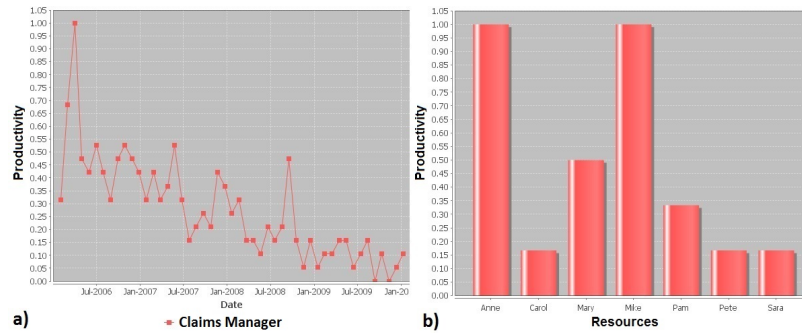


Fig. 6. Examples of outputs for resource productivity evaluation.

4. VALIDATION

This section presents two types of validation for our framework. We first apply our framework to analyse the behaviours of employees in an Australian company. The case study demonstrates real-life examples of the analyses that can be performed by the three modules of our framework (Sections 4.1.1-4.1.3) followed by a discussion of the insights gained from the case study (Section 4.1.4). We then present the results of an online survey conducted with managers to evaluate their opinions about the usefulness of the framework. The framework has been implemented as a plug-in⁷ of the ProM⁸ process mining framework as described previously [Pika 2015].

⁷<http://yawlfoundation.org/risk/files/MiningResourceProfiles.7z>

⁸<http://www.promtools.org>

4.1. Analysing Employee Behaviours in an Australian Company

We applied our framework to analyse employee behaviours in an Australian company. The company was interested in analysing the behaviours of 34 selected employees in order to:

- (1) Evaluate the performance of the employees (the company representative already had an idea about the performance levels of different employees and was interested to see if the analysis results would confirm the assumptions);
- (2) Identify opportunities for improvement of their performance; and
- (3) Identify opportunities for improving process performance.

The data was collected from different information systems, anonymised and cleaned up based on input from process experts. It was an iterative process that involved a number of meetings with company representatives during which we discussed the data requirements and agreed on the data attributes used in the analysis. The resulting event log contained 17,750 cases and more than 700,000 events. Events in the log contained five basic attributes (i.e. *caseid*, *task*, *type*, *time* and *resource*) and also attributes *case_type*, *feedback* (customer feedback), and *experience* (the number of days an employee had been employed by the company at the time of event occurrence). The original data contained information about roles and employees involved in events. We created two versions of the event log. In the first version of the log, we only selected role information and associated it with attribute *resource*. This version was used to demonstrate role-based analysis (RBI 2.1 in Section 4.1.1). In the second version of the event log, we only selected employee information and associated it with attribute *resource*. This version was used to analyse employee behaviours (all other analyses in the case study). We analysed weekly behaviours of the 34 employees over a period of two years. The employees were only involved in one process represented in the event log used in the analysis. The company's feedback on the usefulness of the analysis results was collected via an unstructured interview. In the following sub-sections, we first provide examples from the case study illustrating the three modules of our framework and then discuss the insights gained from the case study and show how they can help improve process and employee performance.

4.1.1. Module 1: Analysing Resource Behaviour. We demonstrate examples of RBIs from each category of resource behaviour. An example of an indicator that reflects resource skills is depicted in Figure 7 (*RBI 1.2 "Case types"*). It shows the fractions of completed cases related to five products (referred to here as *A*, *B*, *C*, *D* and *E*) in which employee *R10* was involved with respect to the total number of completed cases in which *R10* was involved. The chart reveals that employee *R10* was typically involved in cases related to products *A* and *B*, but we can see that the employee was not very experienced in cases related to product *E*.

Figure 8 depicts the values of *RBI 2.2 "Number of cases completed in which a resource was involved"* (a utilisation indicator) for employees *R4* and *R5* who play the same role. We can observe an upward trend for employee *R5*: the number of cases in which *R5* was typically involved gradually increased from 25 cases per week in January 2013 to 50 cases in July 2014, while the number of cases in which *R4* was involved was stable starting from July 2013 at around 17 cases per week. Having analysed this utilisation indicator for the 34 employees we could observe three types of behaviour: some employees were becoming more active over time, other employees displayed stable behaviour and some employees showed irregular patterns being more or less active during different periods. Figure 9 depicts another example of a utilisation indicator (*RBI 2.1*) for three roles, referred to here as *Role 1*, *Role 2* and *Role 3*. While the num-

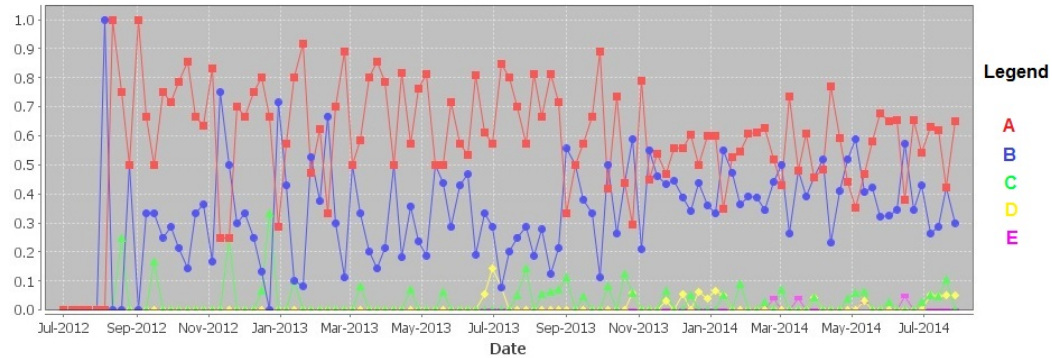


Fig. 7. RBI 1.2 “Case types” for employee R10: Fractions of cases related to five products.

bers of task instances completed during the week by *Role 2* and *Role 3* did not change significantly, employees in *Role 1* tended to complete more task instances over time.

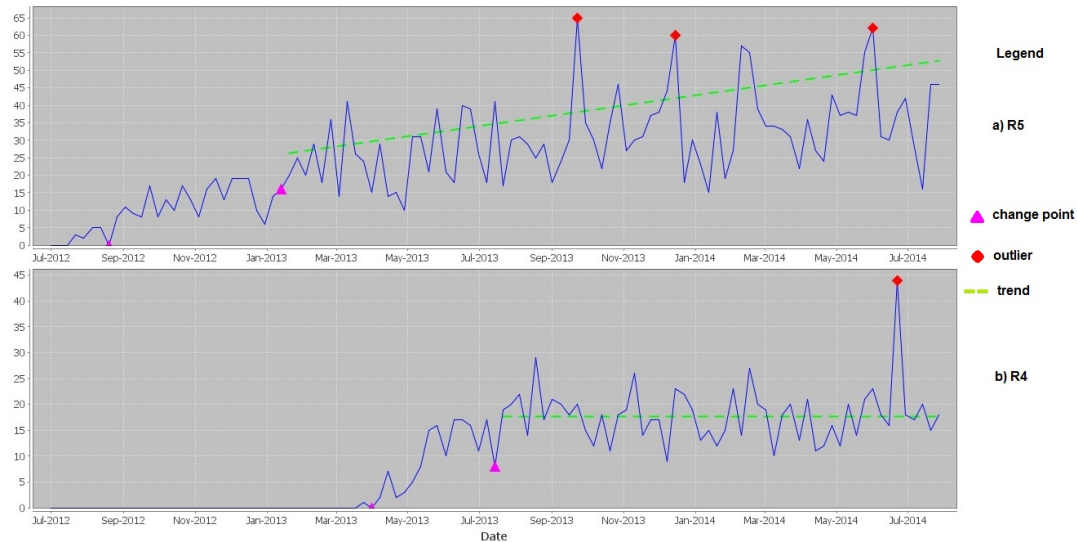


Fig. 8. RBI 2.2: Number of cases completed per week in which resource (a) R5 and (b) R4 were involved.

An example of an indicator that illustrates a resource’s preferences is depicted in Figure 10. It shows the number of task reassignments from employees *R3* and *R15* (RBI 3.3 “Activity reassignments”). As we can see, starting from May 2013, the number of times when a task assigned to employee *R3* was completed by another employee increased and during some weeks it was around 70. The number of task reassignments from employee *R15* also increased from May 2013 but it typically did not exceed 10. We could observe significant differences in the number of task reassignments for different employees having the same role. While for some of them, the high number of task reassignments can be explained by a high workload, the reasons for frequent task reassignments from other employees should be investigated.

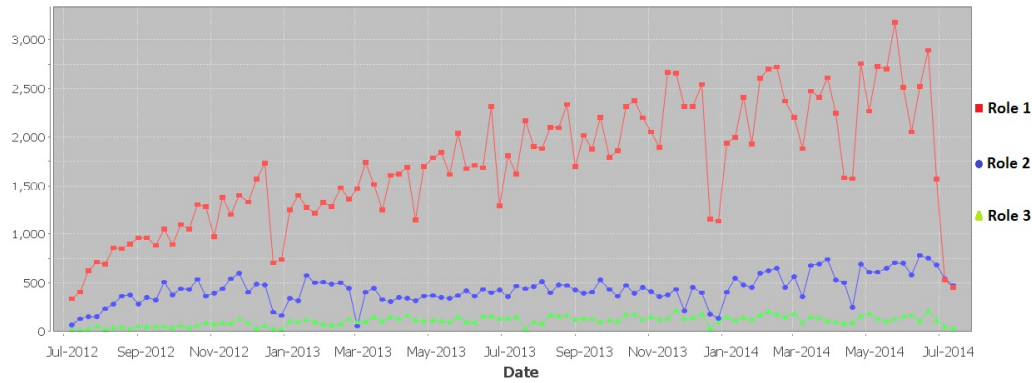


Fig. 9. RBI 2.1: Number of task instances completed per week by employees in *Role 1*, *Role 2* and *Role 3*.

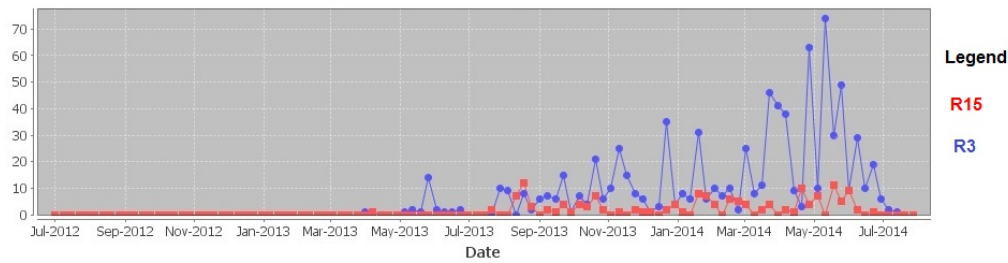


Fig. 10. RBI 3.3: Number of task reassignments from employees R15 and R3.

The values of *RBI 4.3* “Average duration of a given activity” (a productivity indicator) for employees *R16*, *R17* and *R18* are depicted in Figure 11. We can clearly see that employee *R17* was much faster in the execution of this activity (the average duration was typically below 100 hours) than the two other employees for whom the average duration of the task sometimes reached more than 500 hours. We could observe similar differences for other tasks and employees. The company will explore ways of improving process performance by creating groups of efficient employees for each task. Another example of a productivity indicator is depicted in Figure 12, showing *RBI 4.5* “Average customer feedback” for cases managed by employee *R11*. The values of this indicator are based on a sample of cases rated by customers. The indicator values ranged between three and four during most weeks, which means that the quality of service was rated by customers “as expected” (value three) or “higher than expected” (value four) for cases managed by *R11* (value zero means that no cases were completed during the week which were rated by customers and managed by *R11*). During some weeks, the average customer feedback was “much higher than expected” (value five).

Figure 13 depicts the values of *RBI 5.2* “Social position” for employees *R7* and *R11*. It is an example of a collaboration indicator and it is the fraction of resources active during a week who were involved in the same cases with a given resource with respect to the total number of resources active during the week. We can see in Figure 13 that the values of the indicator were stable for employee *R7* but they gradually increased for employee *R11* until June 2014 which means that employee *R11* was becoming involved with more employees in the same cases over time.

In order to obtain a more complete picture of a resource’s behaviour it is necessary to look at the resource’s profile. As an example, Figure 14 depicts a profile of employee *R30* that comprises five RBIs from different categories of resource behaviour. We can

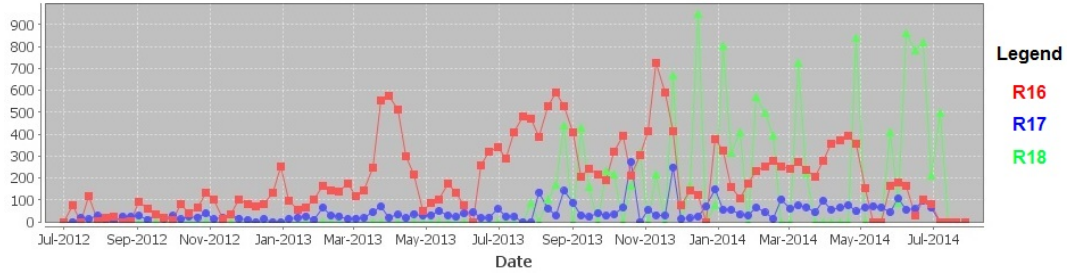


Fig. 11. RBI 4.3: Average duration of a given task completed by an employee.

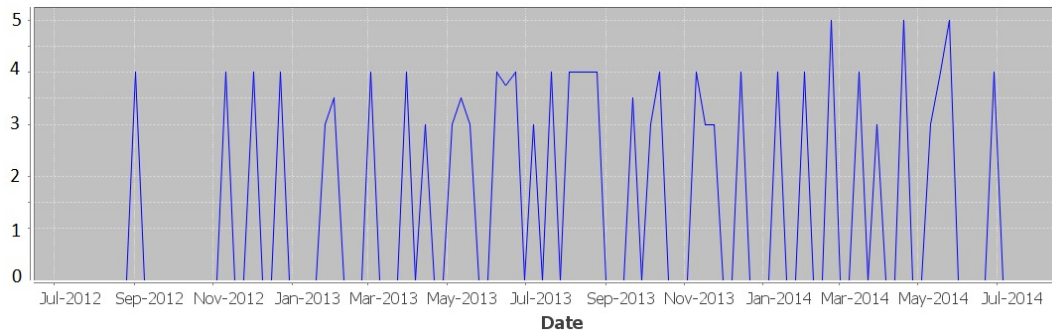


Fig. 12. RBI 4.5: Average customer feedback for cases managed by *R11*.

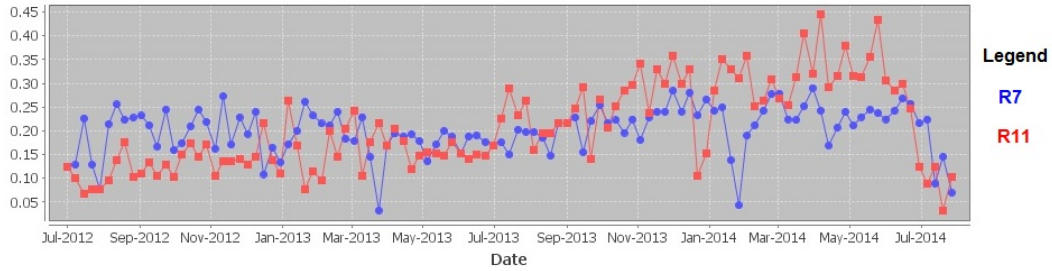


Fig. 13. RBI 5.2: Fraction of employees during a week involved in the same cases with an employee.

observe that there was a change in the employee's behaviour around March 2014. After this point in time the employee performed fewer distinct tasks (Figure 14a) and was involved in more cases (Figure 14b) of shorter duration on average (Figure 14d). The fraction of employees involved in the same cases with *R30* increased (Figure 14e). We can also observe that the number of task reassignments from employee *R30* was typically not very high and that tasks were not reassigned from *R30* before February 2013 (Figure 14c). The changes in resource behaviour we can observe in Figures 14a, b, d and e are related to changes in the employee's set of responsibilities in the organisation.

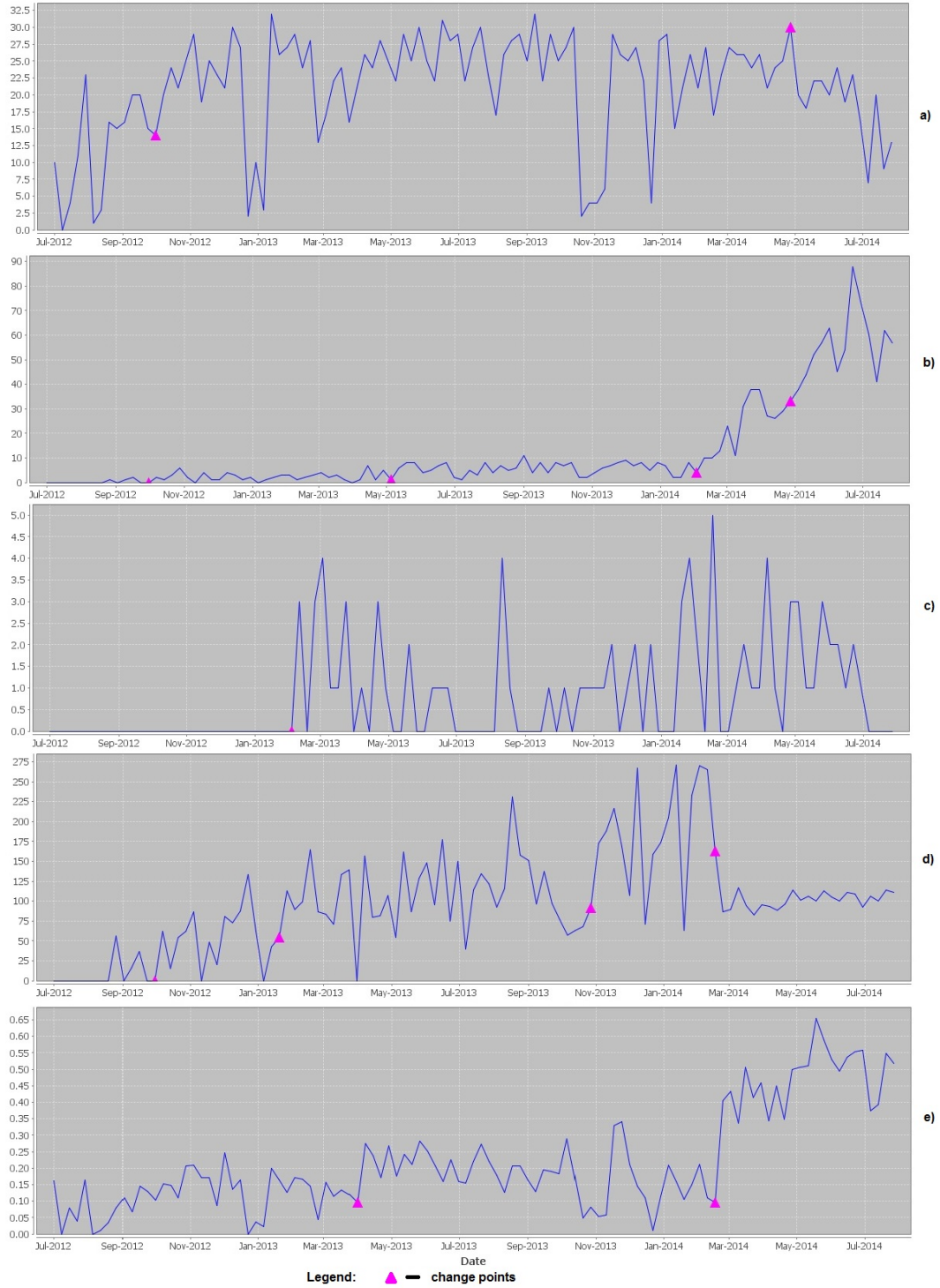


Fig. 14. Profile of employee *R30*. a) *RBI 1.1*: Number of distinct tasks completed, b) *RBI 2.2*: Number of cases in which employee *R30* was involved, c) *RBI 3.3*: Number of task reassignments from *R30*, d) *RBI 4.4*: Average duration of cases in which *R30* was involved, e) *RBI 5.2*: Fraction of employees involved in the same cases with *R30*.

4.1.2. Module 2: Quantifying the Outcome of Resource Behaviour. An example of a process outcome that is available in a typical event log is duration. We consider duration from three perspectives, namely, case, task and time, and show examples from the case study for each perspective. Based on our knowledge of the process and the data, we did not identify any confounding factors that had to be eliminated in the procedure of performing the regression analysis.

We first looked at the *case* perspective and checked the relationship between the number of employees involved in a case (independent variable) and the case duration (dependent variable). All the cases were included in the analysis. In our previous research we found that, in some processes, higher numbers of resources being involved in a case is often associated with long-running cases [Pika et al. 2013a]. We discovered for this process that there is only a weak relationship between the two variables (R^2 is 0.24, $p\text{-value} < .001$); the data and the fitted regression line are depicted in Figure 15a.

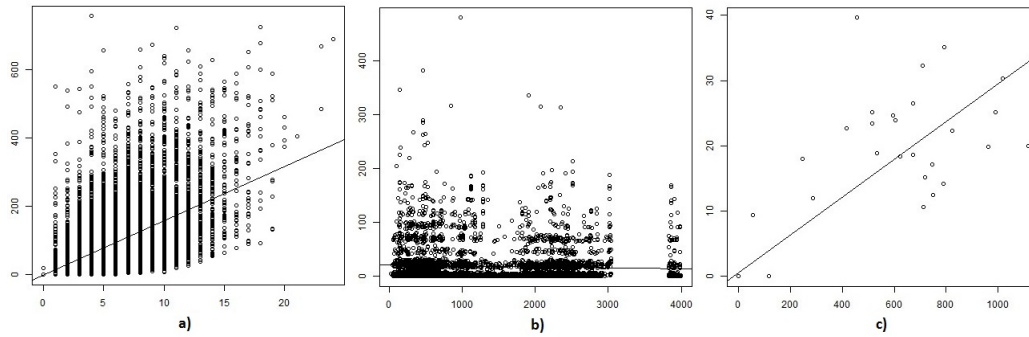


Fig. 15. a) Relationship between the number of employees involved in a case ($x\text{-axis}$) and the case duration ($y\text{-axis}$); b) Relationship between experience of an employee executing a task ($x\text{-axis}$) and the task duration ($y\text{-axis}$); c) Relationship between the number of tasks completed by employee *R5* per week ($x\text{-axis}$) and average duration of the tasks ($y\text{-axis}$).

We then looked at the *task* perspective and checked the relationship between the duration of a task (dependent variable) and the experience of the employee who completed the task (independent variable). Instances of all tasks were included in the analysis. Experience was estimated as the number of days the employee had been employed by the company at the time of task completion. We found that employee experience had no influence on the tasks' duration (R^2 is 0.002, $p\text{-value} < .001$) (Figure 15b).

We also looked at the *time* perspective and checked how the number of tasks completed by an employee during a week (independent variable) affected the average duration of the tasks (dependent variable) for employees *R5* and *R7*. In both cases, workload affects task duration; however, the effect was strong for employee *R5* (R^2 is 0.79, $p\text{-value} < .001$) (Figure 15c) and weak for employee *R7* (R^2 is 0.23, $p\text{-value} < .001$). Based on these observations, the company will explore ways to balance the workload of employees whose efficiency suffers when they are overloaded.

4.1.3. Module 3: Evaluating Resource Productivity. We calculated weekly productivity scores for the 34 selected employees for a period of 109 weeks. We considered as an output the number of executions of a given activity. The company representative assigned the level of complexity to each activity (either high, medium or low) which we used to assign output costs (either five, three, or one, respectively). We did not use inputs as the employees had access to the same resources needed to perform their work

(e.g. information). The 34 selected employees belonged to five different roles and we learned from data the optimal revenue output (i.e. the estimated efficient frontier considering the costs of outputs (Section 3.4)) separately for each role. Then, the weekly productivity scores for each employee were calculated as the division of the observed revenue output of an employee by the optimal revenue output calculated for the employee's role. The resulting productivity scores range from zero to one, with one being the maximum possible productivity⁹.

As an example, Figure 16 depicts the productivity scores for two employees in the same role. We can observe that the productivity of employee *R5* was between 0.4 and 0.6 most of the time and it gradually decreased from March 2014. The productivity of employee *R9* was typically lower than the productivity of *R5* and it varied during different periods. Having extracted the productivity scores for the 34 employees we could observe that some employees in the same role had stable and similar productivity scores while others displayed irregular productivity patterns. In some instances, the productivity scores extracted from the data supported the beliefs of the company representative about the employees' productivity, while the productivity scores of other employees were not as expected (i.e. the scores were higher or lower than expected).

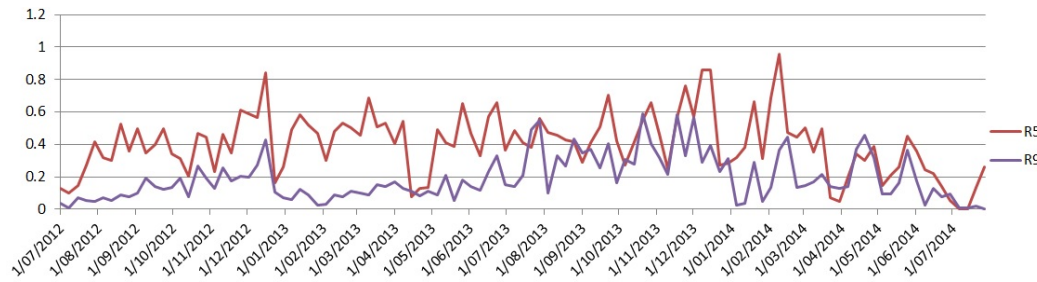


Fig. 16. Weekly productivity scores for employees *R5* and *R9*.

4.1.4. Summary of the Case Study. This section summarises our findings from the case study and discusses the lessons learned. The first goal was to perform an evidence-based evaluation of the performance of the selected 34 employees. The company was interested to see if our findings confirmed their assumptions. We extracted indicators in the five categories of resource behaviour and evaluated the overall productivity of the employees. In most cases, the analysis results confirmed the assumptions identifying better and poorer performers. However, we found that some employees who were suspected of inconsistent performance actually displayed stable behavioural patterns (e.g. in terms of the number of cases and tasks completed and the average duration of the tasks). From the analysis of the behaviours of employees in this process, we could see that abrupt significant changes in employee behaviours were typically related to changes in role or changes in portfolio, outliers were often related to periods of vacations, while gradual changes in behaviours often required attention as they indicated changes in employee performance. We also discovered that some employees were very active (e.g. involved in many cases and tasks); however, they became slower in the execution of some tasks over time, tasks were frequently reassigned from them or they involved many employees in the cases they managed. This shows us that *due care*

⁹This case study does not fully demonstrate the benefit of using DEA for productivity evaluation as we did not use inputs and the levels of activity complexity were assigned by the company representative.

should be taken with the interpretation of results. Each indicator is an abstraction that highlights a certain aspect of resource behaviour while discarding others. In order to obtain a more complete picture of employee performance it would be necessary to look at different dimensions.

The second goal was to identify opportunities for improvement in the performance of the employees. We learned from the analysis that there were significant differences in the amount of time taken to complete a particular task for different employees; hence, the company will be exploring the possibility of assigning tasks to those employees who are more efficient in the execution of a particular task. We also discovered that for some employees the speed of task processing was significantly affected by their workload. The company plans to explore ways to balance the workload of such employees.

Finally, the third goal was to identify opportunities for improving process performance. On the basis of the results, it is expected that case throughput times may be reduced by defining pools of efficient employees for each task and assigning tasks to efficient resources. Balancing employee workload should also contribute to shorter case processing times. We also identified differences in the number of employees processing cases handled by different case managers. The company will be looking into these cases in more detail in order to identify the reasons and suggest best practices for case resourcing.

Applying the framework to analyse the behaviours of employees in an Australian company we *demonstrated the different types of analysis* that the framework allows us to perform. We showed how the framework can *help to evaluate resource performance in a more objective way* and to *identify opportunities for improving the performance of employees and the business process.*

4.2. Survey on the Usefulness of the Framework

We conducted an online anonymous survey to evaluate the perceived usefulness of our proposed framework among managers (as they are the intended users of the framework). The survey described the functionality of each module and showed screenshots of examples of its application using a real data set¹⁰ (the survey questions are available at the yawlfoundation.org site¹¹). Questions were then asked regarding usefulness of the framework features, current status of analyzing resource behaviour in respondents' organisations and process orientation (managers with higher process orientation are expected to see a higher usability of resource performance analyses [Forsberg et al. 1999]).

The questionnaires were distributed using professional social media (e.g. LinkedIn¹²). They were also sent via email to employees in the financial services industry in Germany using a non-public university database of industry contacts available to a member of the research team (the database includes contacts of 2,286 employees, of whom 34.4 % (786) are expected to have managerial positions [Leyer and Moormann 2014]). We asked employees with managerial functions to fill out the questionnaire anonymously. The questionnaire was available in English and in German. In total, 42 managers answered the questionnaire fully (11 in English, 31 in German). The participants had a professional experience of 10.9 years on average (SD: 7.56) and supervised 31.9 (SD: 55.27) employees on average. There was no statistically significant difference between the German and English survey responses; thus, the answers were analysed jointly. Among the respondents, 59.52 % indicated that they worked in the finance and banking industry; others specified the following industries: information

¹⁰The survey did not use the dataset from the case study company.

¹¹<http://yawlfoundation.org/risk/files/Survey.pdf>

¹²www.linkedin.com

Table I. As-is vs. would be helpful with regard to resource performance evaluation

	As-is (M, SD)	Would be helpful (M, SD)
Resource skills	2.88, 0.633	5.78, 0.909
Resource utilisation	3.12, 0.739	5.80, 1.056
Resource preferences	2.59, 0.670	5.57, 1.151
Resource productivity	2.91, 0.726	6.02, 0.869
Resource collaboration	2.36, 0.533	5.45, 1.214

Legend: **M**–Mean; **SD**–Standard deviation; “**As-is**” scale: 1–4 (“Please indicate if [resource behaviour category name] is currently analysed in your organisation when evaluating resource performance”: (1) no opinion, (2) not analysed, (3) analysed based on personal judgements, (4) analysed using information provided by information systems); “**Would be helpful**”: 7-point Likert scale.

Table II. Perceived usefulness of the framework and its modules

Module	(Mean, Standard Deviation)
Framework	(5.71, 1.031)
Analysing resource behaviour	(5.91, 0.878)
Quantifying the Outcome of Resource Behaviour	(5.83, 0.824)
Evaluating resource productivity	(5.86, 1.026)

(7-point Likert scale)

technology, consulting services, manufacturing, education, health, and the restaurant business.

The participants reported a limited application of resource behaviour analysis in the five categories (the “As-is” column in Table I), but would find such analyses to be very helpful (the “Would be helpful” column in Table I). Overall, the modules (including features within the modules) were considered as very relevant by the participants (Table II). The most relevant features (with the percentage of respondents who selected the feature shown in brackets) are: analysing indicators of resource behaviour in the categories of productivity (71.4%), skills (66.7%) and utilisation (61.9%); estimating trends (52.4%); comparing the overall resource productivity scores for multiple resources during different time periods (52.4%) and finding correlations between resource behaviours and outcomes for the *time perspective* (50.0%).

There is a weak statistically significant influence of process orientation on the evaluation of the modules (ns – non-significant): *Analysing Resource Behaviour* (ns (0.427), $R^2 = -.020$), *Quantifying the Outcome of Resource Behaviour* ($p < .05$ (2.039), $R^2 = .072$), *Evaluating Resource Productivity* ($p < .05$ (2.227), $R^2 = .088$), *overall evaluation of the framework* (ns (1.942), $R^2 = .065$). As a consequence, it can be stated that the degree to which the framework was considered relevant by a manager is almost independent of the degree of process orientation of their environment. In addition, the number of subordinates and length of working experience did not have any statistically significant influence on the degree to which the framework was considered relevant.

The survey revealed that *only a limited amount of resource analysis is conducted in companies*. It confirmed that *managers find our framework useful* for resource performance evaluation and that *their opinion of potential usefulness of the framework* is independent of the level of process orientation of their environments, their length of experience or the number of subordinates.

5. DISCUSSION

In this section we first discuss the contributions of our work and compare it with existing approaches. We then summarise assumptions and limitations of our framework and discuss the ways in which our work can be extended.

5.1. Contributions

Human performance measurement and planning are challenging tasks [Espinilla et al. 2013; Peretz and Fried 2012] as human behaviour is complex [Leftwich 2015] and changes over time [Beheshti et al. 2016]. Despite the important role human resources play in business processes [Rosemann and vom Brocke 2015] and the increase of knowledge-intensive tasks in modern organisations [Ciccio et al. 2015; Freel 2016; Wirtz and Lovelock 2016], the BPM community paid little attention to collaborative aspects of BPM [Ciccio et al. 2015]. On the other hand, traditional resource performance evaluation approaches proposed in the management literature are criticised for lack of objectivity [Espinilla et al. 2013] and narrow focus [Neely et al. 2000]. In this article we presented a framework for *mining resource profiles from process execution data*. Knowledge about resource behaviours that can be extracted by our framework *enables evidence-based performance evaluation*, may provide *insights for better workload planning, effective human resource development strategies, and resource and process performance improvement*, as we demonstrated in Section 4.

The framework presented in this article is an extension of our earlier work [Pika et al. 2014] in which we proposed a method for analysing resource behaviour. Here we presented an extended framework that also includes a new method for quantifying the outcome of resource behaviour (Section 3.3) and a new method for evaluating and comparing resource productivity (Section 3.4). We presented the results of a new case study and a survey evaluating the usefulness of our framework (Section 4).

Some technical aspects of our framework were inspired by existing process mining approaches [Pika et al. 2013a; Pika et al. 2013b; de Leoni et al. 2014; van der Aalst 2013; Bose et al. 2013]: we use indicators extracted from event log data; our framework allows the correlation of resource behaviours and outcomes; it allows analysing resource behaviour from different perspectives and it considers changes in resource behaviour over time; however, unlike the above-mentioned approaches, the *focus of our framework is resource behaviour analysis*, rather than business processes.

We adopted DEA [Bogetoft and Otto 2011] for resource productivity evaluation. DEA was previously used to evaluate productivity of employees [Manoharan et al. 2009; Wagner et al. 2003; Koch-Rogge et al. 2014]. However, in these case studies data was collected from different sources and manually transformed into a form suitable for DEA analysis. Our goal, on the other hand, was not to demonstrate the use of DEA in a particular company, but to present *a method that allows us to define resource inputs and outputs, extract their values from an event log and evaluate resource productivity*.

Existing organisational mining approaches extract organisational structures [Song and van der Aalst 2008; van der Aalst et al. 2005], devise resource allocation mechanisms [Liu et al. 2012; Cabanillas et al. 2013; Kumar et al. 2013; Ly et al. 2006] or focus on few specific resource behaviours [Huang et al. 2012; Nakatumba and van der Aalst 2010]. In contrast to these approaches, in this article we presented a framework that *focuses on employees, allows analysing their behaviours from different perspectives, is extensible (allows definition of new measures) and considers the evolution of resource behaviours over time*.

5.2. Assumptions, Limitations and Future Work

We assume that the data recorded in an event log accurately represents a company's business process. If the event log contains incomplete or inaccurate information (e.g. some events or attributes are not recorded or the timestamps of events do not represent the time of event occurrence), then the analysis results may be incorrect. We assume that resources are involved in one process only; otherwise, logs from different processes should be merged ensuring that the case identifiers in the combined log are unique.

Alternatively, the proportion of time a resource spent on the process under analysis should be known to ensure fair comparisons across resources.

We provided examples of measures of resource behaviour in each of the three modules that can be mined from typical event logs. The option of defining more comprehensive sets of resource behaviour measures relevant in different contexts can be explored.

We evaluated the usefulness of our framework by showing examples of its application and conducting an online survey among managers. We also conducted a real case study using a dataset containing the events from a core business process in an Australian company recorded over a two year period. We collected feedback from the case study company via an unstructured interview. A manager from the case study company was not asked to participate in the online survey. To evaluate ease of use of the corresponding software artefact a study can be conducted in which users gain hands-on experience with the tool and provide their feedback.

The framework is based on the analysis of event log data; hence, it can only provide knowledge about the process-related behaviours of resources. Such knowledge can potentially be used in an inappropriate way; for example, a narrow set of resource behaviours may be used to instigate disciplinary action. The possibility of eliminating potential framework misuse should be further investigated. Countries may have different regulations regarding privacy and personal data protection. For example, an employer may need to inform employees about data collection and analysis [van der Aalst et al. 2005] or permission of the relevant employee union may be required (as is the case in Germany). A direction for future work is providing users with an option to select types of resource analysis depending on the regulations that exist.

6. CONCLUSIONS

Managers need objective information about the different working behaviours of resources in order to better understand areas of performance improvement and to make more informed resource-related decisions. Such knowledge may enable better workload planning, improved performance appraisal systems and more efficient resource development strategies and provide insights for resource and process performance improvement.

The framework we presented in this article contributes to the process mining field, specifically to the organisational mining area. In contrast to prior approaches, our framework allows users to extract knowledge about different aspects of the behaviour of individual employees and teams, either using pre-defined indicators or defining their own measures, to investigate the impact of given resource behaviours on process outcomes, evaluate resource productivity, and understand changes in resource behaviour over time.

We performed two types of evaluation. We conducted a case study applying our framework to an event log from an Australian company to analyse employee behaviours in the company. We demonstrated how the framework can be used to gain insights into performance improvement. We also conducted an online survey of managers in order to evaluate the perceived usefulness of the framework and its modules. We found that most respondents did not evaluate many aspects of resource behaviour or they did it solely based on their subjective judgements. They stated that the methods provided by our framework would be useful and relevant to them.

ACKNOWLEDGMENTS

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for conducting the survey was obtained (Approval number: 1400000420). We also used an anonymised data set in our case study, so this component of the research was exempt from ethical review in conformity with the National Statement on Ethical Conduct in Human Research (2007)¹³. We would like to thank the process experts from the case study company for their help and insights shared with us during the case study.

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¹³<https://www.nhmrc.gov.au/guidelines-publications/e72>

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